NN-H-0081b
September 16, 1957
Interim Revision of
Fed. Spec. NN-H-81a
December 4, 1950

INTERIM FEDERAL SPECIFICATION

HANDLES, ASH

This Interim Federal Specification is based upon currently available technical information, but has not been approved for promulgation as a regular Federal Specification. It is subject to modification and pending its promulgation as a regular Federal Specification, may be used in procurement.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers ash handles for use in post hole diggers, forks, hoes, rakes, and shovels as described in the current Federal Specifications GGG-D-326, GGG-F-601, GGG-F-626, GGG-H-496, GGG-R-96, and GGG-S-326. (See 2.1).

1.2 Classification. - The handles shall be of the types, classes, styles and lengths as specified to fit the tools referenced in the appropriate Federal specifications.

1.3 Grades. - The handles shall be of the following grades, as specified. (See table I).

Grade A. - For forks, hoes, and rakes.
Grade SA. - For shovels.
Grade B. - For forks, hoes and rakes.
Grade SB. - For shovels and post-hole diggers.

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 The following Federal Specifications and Military Standards of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

QQ-S-781 - Strapping, Flat; Steel
GGG-D-326 - Diggers, Post-Hole
GGG-F-601 - Forks; Hay, Manure, Mill and Spading
GGG-F-626 - Fork, Stone, Hand, D-Handle
GGG-H-496 - Hoes, Garden and Mortar
GGG-R-96 - Rakes, Hand
GGG-S-326 - Shovels, Scoops, Spades, and Telegraph Spoons

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following publications of the issues in effect on date of invitation for bids form a part of this specification.

Consolidated Classification Committee

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois.)

Department of Commerce

Simplified Practice Recommendation R-76-40 - Ash Handles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Material. - The handles shall be white ash which includes wood cut from white ash (Fraxinus Americana), green ash (Fraxinus Pennsylvanica) and blue ash (Fraxinus quadrangulata).

3.2 Moisture content. - The moisture content of the handles shall not exceed 12 percent of the oven-dry weight.

3.3 Quality. - The handles shall conform to the requirements of Table I for the grades specified.

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3.4 Size and form. - The handles shall be formed and shaped to fit properly the tools referenced in the invitation for bids and described in the appropriate Federal Specification and shall be of the diameters and lengths given therein, as specified.

3.5 Workmanship. - The handles shall be smoothly finished and shall be of good commercial workmanship.

3.6 Rivets. - When replacement handles which are customarily fastened to the tools with rivets are ordered, two suitable oval-head rivets shall be supplied with each handle. The rivets shall be fastened to the handle in a manner to preclude their being lost during shipment.

3.7 Marking. - Handles shall be marked in a plain and permanent manner with the manufacturer's name or with a trade mark of such known character that the source of manufacture may be readily determined.

4. SAMPLING, INSPECTION AND TEST PROCEDURE

4.1 QUALITY ASSURANCE PROVISIONS (For the use of the Defense Agencies)

4.1.1 Sampling for lot acceptance.

4.1.1.1 Inspection lot. - All handles of the same type, class, style design, grade, and length offered for delivery at one time shall be a lot for purpose of inspection and test.

4.1.1.2 Sampling for examination. - A random sample of handles shall be selected from each lot offered for Government inspection in accordance with Military Standard MIL-STD-105 at inspection level III for lots up to and including 40, inspection level II for lots of 41 to 300, and inspection level I for lots over 300. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.1.1.3 Sampling for lot acceptance tests.

4.1.1.3.1 Sampling for moisture content test — electrical moisture meter method. - A random sample of handles shall be selected from each lot offered for Government testing in accordance with the appendix to Military Standard MIL-STD-105 at inspection level L-7. The AQL shall equal 4.0 percent defective. However, the sample size shall be the number associated with the letter in the table in the appendix, and the acceptance number shall be zero until the AQL permits one or greater.

4.1.1.3.2 Sampling for other tests. - A random sample of handles shall be selected from each lot offered for Government tests in accordance with the appendix to Military Standard MIL-STD-105 at inspection level L-4.

4.1.2 Inspection.

4.1.2.1 Visual and dimensional examination. - Each of the sample handles selected in accordance with 4.1.1.2 shall be visually and dimensionally examined by the

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Government inspector to verify compliance with this specification. Any handle in the sample containing one or more visual or dimensional defects, shall be rejected, and if the number of defective handles in any sample exceeds the acceptance number for that sample, the lot represented shall be rejected.

4.1.3 Tests.

4.1.3.1 Moisture content test — electrical moisture meter method (see 4.4.2). - Each of the sample handles selected in accordance with 4.1.1.3.1, shall be subjected to the test specified in 4.1.4.1.2. Any handle in the sample which fails the test shall be rejected and if the number of defective handles in any sample exceeds the acceptance number for that sample, the lot represented shall be rejected.

4.1.3.2 Other tests. - Each of the sample handles selected in accordance with 4.1.1.3.2 shall be subjected to the tests specified in 4.1.4.1 and 4.1.4.1.1. If any handle fails any test, the lot shall be rejected.

4.1.4 Test procedures.

4.1.4.1 Moisture content. - Each of the sample handles selected shall be tested to determine conformance with the moisture content requirements of 3.2 by the oven drying method or by the electrical moisture meter method if approved by the bureau or agency concerned.

4.1.4.1.1 Oven drying method. - A sample section, 1 inch long, shall be cut not less than 6 inches from either end of the handle. Immediately after sawing, all loose splinters shall be removed, and the sample weighed on a scale that is accurate to 0.5 of 1 percent. (It is important that the weight be taken immediately after sawing, since the wood is subject to moisture changes on exposure to air.) The sample shall be dried in an oven maintained at a temperature of 212° Fahrenheit (F.) to 221°F. (100° Centigrade (C.) to 105°C.) until a constant weight is attained. At the end of this period, the sample shall be reweighed; this being the oven dry weight. The percentage of moisture content shall be computed by the following formula:

$$\frac{\text{Original weight} - \text{Oven dry weight}}{\text{Oven dry weight}} \times 100 = \text{Moisture content in percent.}$$

4.1.4.1.2 Electrical moisture meter method. - The moisture content shall be determined in accordance with instructions furnished by the manufacturer of the electrical moisture meter. Electrical moisture meters shall not be used on handles that are surface-wet or that have been color-stained. In determining the moisture content of the handles, the reading on the electrical moisture meter shall be taken when the meter electrodes are driven to the center of the handles being tested.

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4.2 SAMPLING, INSPECTION, AND TEST PROCEDURES. (For the Use of Civilian agencies.)

4.2.1 Sampling for lot acceptance.

4.2.1.1 Inspection lot. - All handles of the same type, class, style design, grade, and length offered for delivery at one time shall be a lot for the purpose of inspection or test.

4.2.1.2 Sampling for examination. - A random sample of handles shall be selected from each lot offered for visual and dimensional examination. The size of sample and acceptance and rejection numbers shall be as indicated in Part A of the Sampling Table.

4.2.1.3 Sampling for moisture content test — electric moisture meter method. - A random sample of handles shall be selected from each lot offered for electrical moisture meter test. The size of sample and acceptance and rejection numbers shall be as indicated in Part B of the Sampling Table.

4.2.1.4 Sampling for destructive tests. - A random sample of handles shall be selected from each lot of 41 or more handles offered for destructive tests as specified in paragraph 4.2.4. The size of sample and acceptance and rejection numbers shall be as indicated in Part C of the Sampling Table.

4.2.2 Visual and dimensional examination. - Each of the sample handles selected in accordance with paragraph 4.2.1.2, shall be visually and dimensionally examined by the Government inspector to verify compliance with this specification. Any handle in the sample containing one or more visual or dimensional defects, shall be rejected, and if the number of defective handles in any sample exceeds the acceptance number for that sample, the lot represented shall be rejected.

4.2.3 Electrical moisture meter test. - Each of the sample handles selected in accordance with paragraph 4.2.1.3, shall be tested for moisture content by the electrical moisture-meter method. The moisture test shall be in accordance with instructions furnished by the manufacturer of the electrical moisture meter. Electrical moisture meters shall not be used where the surface of the handle is wet or has been color-stained. The reading on the electrical moisture meter shall be taken when the meter electrodes are driven to a depth of one sixth the thickness of the handle being tested. Any handle showing more than 12 percent moisture content shall be considered defective, and if the number of defective handles in the sample exceeds the acceptance number for that sample, the lot represented shall be rejected.

4.2.4 Destructive tests. - Each of the sample handles selected in accordance with paragraph 4.2.1.4 shall be subjected to destructive tests as specified in paragraphs 4.2.4.1 and 4.2.4.2. Any handle failing either test shall be considered defective, and if the number of defective handles in the sample exceeds the acceptance number for that sample, the lot represented shall be rejected.

4.2.4.1 Moisture content — ovendrying method. - (To be used when a moisture meter is not available). — A sample section, 1 inch long, shall be cut not less than

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6 inches from either end of the handle. Immediately after sawing, all loose splinters shall be removed, and the sample weighed on a scale that is accurate to 0.5 percent. (It is important that the weight be taken immediately after sawing, since the wood is subject to moisture changes on exposure to air.) The sample shall be dried in an oven maintained at a temperature of 217° ± 4° Fahrenheit (103° ± 2° Centigrade) until a constant weight is attained. At the end of this period, the sample shall be reweighed; this being the oven-dry weight. The percentage of moisture content shall be computed by the following formula:

$$\frac{\text{Original weight — oven-dry weight}}{\text{Oven-dry weight}} \times 100 = \text{Moisture content in percent}$$

4.2.4.2 Weight per cubic foot. - A sample section not less than 1 inch in length cut as specified in paragraph 4.2.4.2, shall be used for a determination of weight per cubic foot. After conditioning to 12 percent moisture content, 1/ the sample shall be weighed to an accuracy of 0.5 percent and shall then be immersed in a hot paraffin bath, care being taken to remove it quickly to insure a thin coating. The volume of the paraffin-coated sample shall be determined by immersion in water, the weight of water displaced by grams being the volume of the sample in cubic centimeters. Weight of the sample in grams divided by volume of the sample in cubic centimeters gives the specific gravity. Specific gravity multiplied by the weight per cubic foot of water (62.4 pounds) gives the weight per cubic foot of the wood.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking (see 6.1)

5.1.1 Level A and Level B.

5.1.1.1 D Handles. - D handles shall be packed in bundles of 12 arranged in two layers of six each. The bundles shall be formed by placing six handles on edge nesting into one another with the "D" grips in alignment at one end. The second layer shall be formed in the same manner on top of the first layer but reversed end-for-end. Four wood cleats, approximately 8-1/4 inches long, 2 inches wide, and 3/4 inch thick, shall be applied to the bundle; two cleats on top and two on the bottom, each cleat approximately 10 inches from each end of the bundle. Each bundle shall be bound with 3/4 x 0.020-inch steel straps conforming to Federal Specification QQ-S-781. One strap shall be applied over each pair of cleats.

5.1.1.2 Straight and bent long handles. - Handles of this description shall be packed in bundles of 24 arranged in four layers of six each. The grips of the handles in the first two layers shall be one end and the handles in the third and fourth layers shall be reversed end-for-end. Each bundle shall be bound with two 3/4 x 0.020-inch

1/ Conditioning for 1 week with ends exposed in an atmosphere at 80°F. and 65 percent relative humidity will result in a moisture content near enough to 12 percent to be acceptable under this provision.

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steel straps conforming to Federal Specification QQ-S-701. One strap shall be applied approximately 10 inches from each end.

5.1.1.3 Marking. - In addition to any special marking required by the contract or order, packages shall be marked in accordance with Standard MIL-STD-129.

5.1.2 Level C. - The handles shall be shipped in accordance with standard commercial practice, so as to insure acceptance by carriers for transportation at the lowest rate to the point of delivery and to give reasonable protection against normal hazards of transportation.

5.1.2.1 Marking. - Unless otherwise specified in the contract or order, each package shall bear a label or tag giving the type, class, style, and grade of handles, the number of the contract or order, and the name of the contractor.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title number, and date of this specification.
(b) Description of handles required by reference to the type, class and style of the tool in which they are used (see 1.2).
(c) Length of handles (see 1.2).
(d) Grade of handles (see 1.3).
(e) Level of packing and marking required (see 5.1).

6.2 Treatment. - Sometimes rough bolts as well as unfinished squares and stored handles are severely damaged by infestations of insects, generally Lyctus powder-post beetles. Bolts cut during the active season are subject to attack by Lyctus beetles as soon as the moisture content is below 20 percent. This is also true of unfinished squares or manufactured but unfinished implement handle stock and similar items. Fortunately Lyctus attack can be prevented by dipping the wood items in an insecticide. To be effective the insecticide must be applied to raw wood since it will not penetrate paints, varnishes, lacquers, waxes or other finishing materials. For those who are interested in insecticide dips, a consideration of Appendix A is suggested. The methods described therein may be specified in connection with the procurement of handles or they may be followed by the purchaser. Lyctus infestations can be very costly. Adequate treatments are relatively inexpensive and should be included as part of the manufacturing process.

6.3 Federal specifications do not include all types, classes, grades, size, etc., of the commodities indicated by the titles of the specification, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished,

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APPENDIX A. - TREATMENT TO PREVENT INFESTATION

11. Treatment. - After fabrication, sanding, marking, and before the installation of ferrules, dip the handles for 3 minutes in a solution of DDT, conforming to Federal Specification O-I-531. Care shall be exercised to prevent DDT solution from coming into contact with those dipping the handles. The handles shall be allowed to drain and dry before shipping or storing. 1/

12. Determination of the presence of DDT. - From a 12-inch section of a treated handle plane off 1/4" and discard the shavings. Take the next fairly thick shaving for submission to a selected laboratory for testing as follows:

12.1 Preparation of reagent. - Seventy-five milliters (ml.) of reagent grade pyridine and 50 grams of potassium hydroxide pellets shall be heated for 1 hour on a steam bath in a liter Erlenmeyer flask, fitted with an all glass reflux condenser. The pyridine shall be cooled and decanted into a second Erlenmeyer flask containing an additional 50 grams of potassium hydroxide and shall then be refluxed on a hot plate for 45 minutes. The pyridine shall be cooled, decanted and distilled in an all glass distillation apparatus. Two-tenths percent water and 0.2 percent zanthydrol shall be added to the distillate.

1/ Green bolts may be protected from Lyctus infestation by giving them a 10 second dip in a 5 percent water emulsion of DDT. Seasoned (10 to 15 percent Moisture content) bolts and unmanufactured squares may be treated as stipulated for the fabricated handles.

12.2 Test procedure. - Scrapings or thin shavings from the handle shall be placed in a beaker and covered with benzene. The benzene shall be brought to a boil and then filtered into a test tube. The benzene solution shall be evaporated to dryness. Five ml. of the pyridine xanthydrol solution and four pellets of potassium hydroxide shall be added and heated to boiling for 3 to 5 minutes. The reagent alone develops a yellow color changing to dark green which on cooling reverts to a light greenish yellow color. When DDT is present a red color develops first, followed by a dark green.

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TABLE I.

The quality requirements given herein conform to Simplified Practice Recommendation R76-40, Ash Handles, for grades A, SA, B, and SB except that weight limitations are based on a moisture content not exceeding 12 percent instead of 8 percent. Long handles and stems for forks, rakes, hoes, and kindred farm and garden tools are described by grades A and B; long handles and stems for shovels, spades, and scoops, by grades SA and SB. A stem is a short handle used to make D handles. The grading of stems conforms substantially to that specified for the lower two-thirds of corresponding long handles.

Grade	Color	Number of annual rings per inch of radius		Weight in pounds per cubic foot[1]	Maximum permissible cross grain[2]		Admissible blemishes	
		Min.	Max.		For stems	For long handles	Kind	Number
A	White, light red, and light brown; dark red and brown in 10 inches at both ends.[3]	5	17	Over 44 (heavy)	1/2 diameter in full length	1/2 diameter in bottom 2/3, 1/2 diameter in top 1/3, 2/3 diameter in top 7 inches	Slight dip grain (not more than 1 per handle), pin twig marks, small streaks, and slight butterfly marks[4]	One for each 24 inches of handle length of fraction thereof, at least 12 inches apart.
SA	White, light red, and light brown; dark red and brown in 10 inches at tool end.	5	15	--do---	--do---	---do---	---do---	Do.

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TABLE I. (Continued)

Grade	Color	Number of annual rings per inch of radius		Weight in pounds per cubic foot[1]	Maximum permissible cross grain[2]		Admissible blemishes	
		Min.	Max.		For stems	For long handles	Kind	Number
B	White, red, brown, and mildly variegated (calico); dark red and dark brown in 10 inches at both ends.[3]	3	21	Over 38 to 44 (medium)	1 in 12 anywhere in the full length	1 in 12, 1 in 6 in 12 inches at one end	Small knots, if located at the ends, and slight dip grain (not more than two of each per handle), pin twig marks medium streaks, medium butterfly marks, and wavy grain[4,5]	Two for each 18 inches of handle length or fraction thereof, at least 6 inches apart.
SB	White, red, and brown dark brown in 10 inches at tool end.	4	18	---do--	Full diameter in any 30 inches of length	Full diameter in any 30 inches of bottom 2/3, 3/4 diameter in top 1/3, 3/4 diameter in top 7 inches	Slight dip grain pin twig marks, medium streaks, medium butterfly marks, and wavy grain[4]	Do.

[1]Based on a moisture content not exceeding 12 percent: [2]Deviation of the fiber from a line parallel with the axis of the handle. For example: 1/2 diameter in bottom 2/3 means that the grain may cross 1/2 the diameter of the handle is not less than 2/3 the length, measured from the bottom: [3]Dark red and dark brown permitted on stems at tool end only: [4]Blemishes are defined and classified as follows: Butterfly mark. - Discoloration resembling somewhat the trail left by a snail; Medium butterfly mark. - In contrast with the color of the wood; Slight butterfly mark. - Matching the general color of the wood; Pin twig mark. - Mark made by a very small twig; Streak. - Thread-like discoloration; Medium streak. - 1 to 4 inches in length; Small streak. - Not more than 1 inch in length; Dip grain (knurl). - Grain dipping into and out of the handle indicated by a fairly oval contour of grain on the surface; Slight dip grain. - Not less than 3 inches long, nor deeper than 1/8 the minimum diameter of the handle at the point where the dip occurs; Small knot. - Average diameter not more than 1/8 inch. Admissible blemishes shall not be such as to impair the strength or serviceability of the handle. [5]Knots will be permitted in stems at tool end only.

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Sampling Table

Lot size	A			B			C		
	Sampling for examination			Sampling for electrical moisture-meter tests			Sampling for destructive tests		
	Sample size	Acceptance number	Rejection number	Sample size	Acceptance number	Rejection number	Sample size	Acceptance number	Rejection number
2 to 15	Lot	0	1	2	0	1	-	-	-
16 to 40	15	0	1	3	0	1	-	-	-
41 to 110	15	0	1	5	0	1	2	0	1
111 to 300	35	1	2	7	0	1	3	0	1
301 to 800	35	1	2	15	1	2	5	0	1
801 to 1300	50	2	3	25	2	3	7	0	1
1301 to 3200	75	3	4	35	3	4	10	1	2
3201 and over	150	5	6	50	4	5	15	1	2

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